

Bringing Standards into Product: ISO 26262 + SOTIF + ASPICE in a L4 Autonomous Transportation System

Yang Xiao (*R&D Engineer*)

Maarten Klaassen (*System Assurance Manager*)

Jeroen Ploeg (*Lead Controls Architect*)

2getthere

September 11, 2026



The claim of this talk



2getthere GRT (Group Rapid Transit) — vehicles, operations control, infrastructure and vehicle in operation

Positioning of 2getthere

- > 2getthere delivers the **ATS (Autonomous Transportation System)** as OEM
- > (30+) years building transit systems
- > Speaking from a **commercial product in operation**, not a research vehicle

Our safety strategy - framework that combines the strengths of three complementary standards:

ISO 26262

Hazard-driven safety lifecycle

- > Requirements organised in tiers
- > Verified architecture

ISO 21448

Absence of risk from functional insufficiency

- > False positive (FP) / False negative (FN) characterisation, functional-insufficiency analysis
- > Residual-risk acceptance via operational evidence

ASPICE (Automotive SPICE)

Process capability & evidence framework

- > Software development process framework
- > ISO 26262-6 layered on top for ASIL (Automotive Safety Integrity Level)-specific technical rigor

Outline

1. Our Level-4 Autonomous Driving (AD) vehicles and products

GRT (Group Rapid Transit) / PRT (Personal Rapid Transit) product family and AD software stack

2. Overview of our safety strategy/process

ISO 26262 + SOTIF (Safety Of The Intended Functionality) + ASPICE

3. ISO 26262 walkthrough — concept, system, hardware

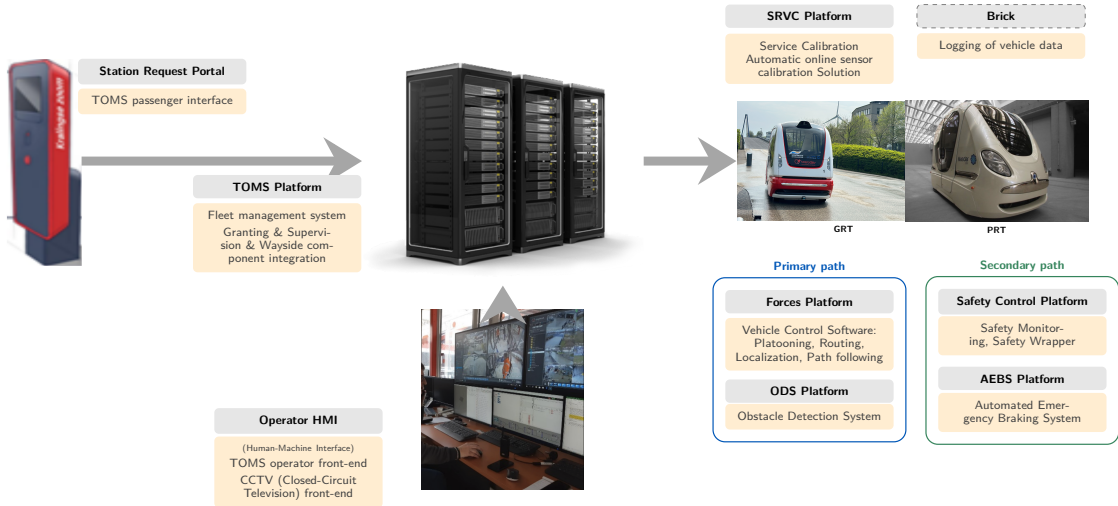
HARA → safety goals → FSC/FSRs → TSC/TSRs → HW analysis

4. Software safety — ISO 26262-6 + ASPICE

5. SOTIF — beyond ISO 26262

6. Conclusions

Software Stack for our Level-4 Autonomous Driving (AD) vehicles



Our Safety Strategy based on Three Elements Framework

From concept to operation

ISO 26262

Hazard-driven E/E product safety lifecycle

- What hazards exist? How serious?
- How does the function avoid them? With what architecture?
- What safety mechanisms are in place?
- Are HW failure rates low enough?
- Is the SW built and verified to ASIL?

ASPICE

SW process capability & engineering evidence (SWE — SoftWare Engineering, 1–6)

- Is process capability evidenced?

SOTIF

Residual risk from functional insufficiency

- Is the function sufficient when nothing fails?
- For collision avoidance system: False Negative (FN) and False Positive (FP) are the two main safety concerns for perception-based collision avoidance systems.
Triggering conditions include adverse weather, sensor interference, retro-reflectors, and sensor pollution.

1 · Definition

Concept & safety definition

- HARA (Hazard Analysis & Risk Assessment)
- Stakeholder Safety Requirements
- Safety Concept
- System safety requirements
- Software safety requirements
- Test case design for safety requirements

2 · Development

Design & safety analysis & software development

- FMEA (Failure Mode and Effects Analysis) – inductive analysis
- Fault Tree Analysis (FTA) — deductive analysis
- Software Architecture, Detailed Design and Code implementation
- Software Test Reports
- V&V Test Strategy
- SOTIF strategy
- Triggering Condition Analysis (TCA)

3 · Qualification

Test & certification

- V&V test reports
- Final Design Safety Case
- Product ASPICE report

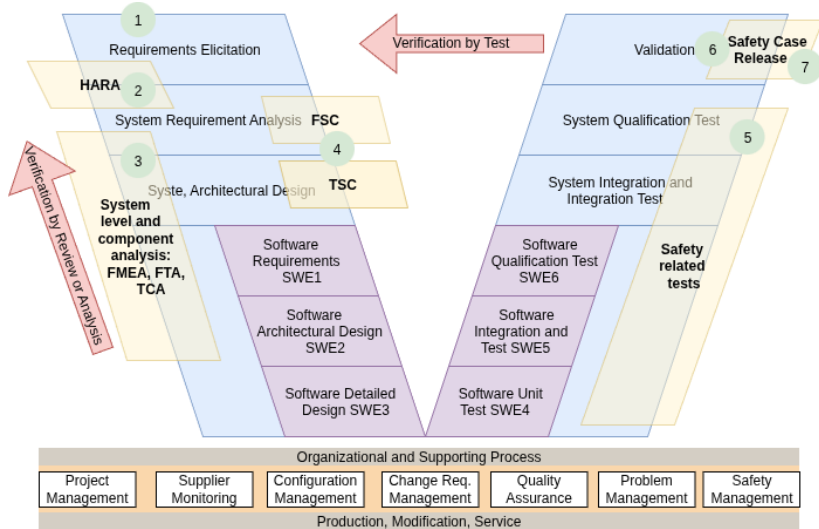
4 · Utilization

Operation

- Operational safety work packages
- In-service hazard monitoring
- SOTIF residual risk monitoring

Our Safety Process Built on the ISO 26262 V-Model

SOTIF supports the design phase, ASPICE supports software development

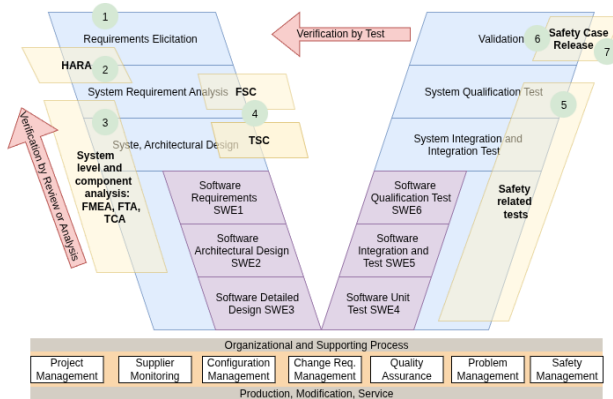


Our Safety Process Built on the ISO 26262 V-Model

SOTIF supports the design phase, ASPICE supports software development

ISO 26262

- **HARA** → **Safety Goals**
- **Functional Safety Concept (FSC):** Functional Safety Requirements (FSRs) and Coarse architecture
- **Technical Safety Concept (TSC):** Technical Safety Requirements (TSRs) and System architecture
- **Hardware & system analysis:** FMEA,FTA
- **Software development (SWE.1–6):** Software Requirements, architecture, detailed design, test reports, SWE1-6 work products.
- **Safety Case & Release:** Final Design Safety Case Report

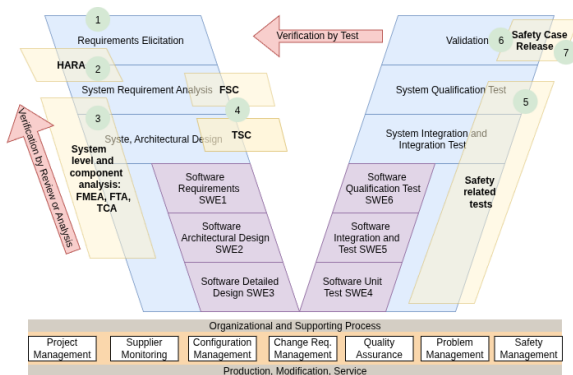


Our Safety Process Built on the ISO 26262 V-Model

SOTIF supports the design phase, ASPICE supports software development

ISO 26262

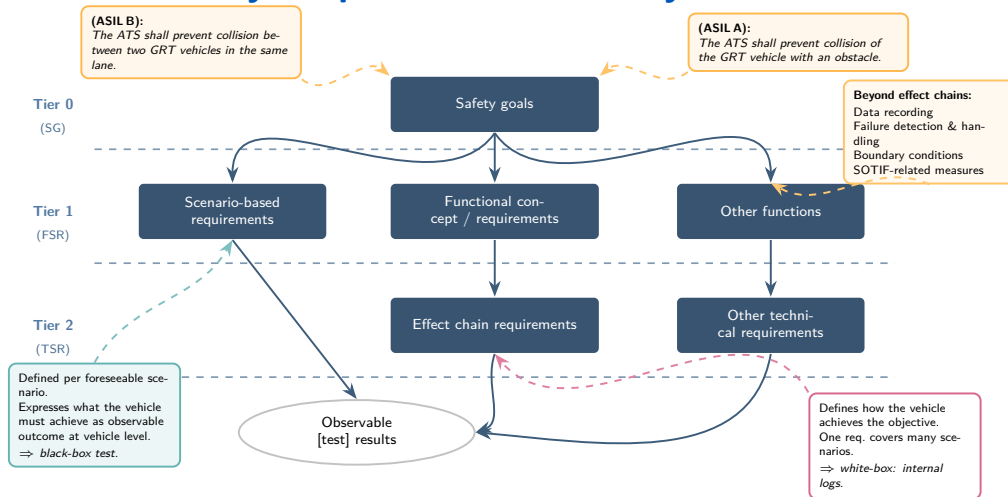
- **HARA** → Safety Goals
- **Functional Safety Concept (FSC)**: Functional Safety Requirements (FSRs) and Coarse architecture
- **Technical Safety Concept (TSC)**: Technical Safety Requirements (TSRs) and System architecture
- **Hardware & system analysis**: FMEA, FTA
- **Software development (SWE.1–6)**: Software Requirements, architecture, detailed design, test reports, SWE1-6 work products.
- **Safety Case & Release**: Final Design Safety Case Report



SOTIF

- **1: Functional & System Specification (Inputs for SOTIF Analysis)**
- **2 & 3: SOTIF HARA and SOTIF Analysis** — TCA, hazard identification; functional insufficiencies; performance limitations; risk evaluation, acceptance criteria
- **4: Specification of SOTIF Measures** — functional modification/improvement; restriction of the function/ODD
- **5: SOTIF V&V Strategy, Evaluation of Hazardous Scenarios**
- **6: SOTIF Release**
- **7: Operation Activities**

Three-Tier Safety Requirement Hierarchy

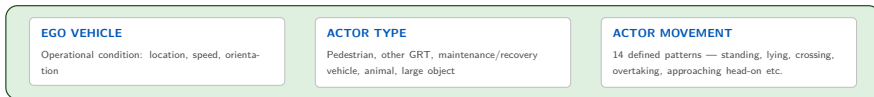


[1] System Safety Team, "Collision Avoidance Safety Concept," 2getthere B.V., Utrecht, The Netherlands, Rep. 6039.06.05.002, v1.1, Apr. 1, 2025.

Functional Safety Concept

Scenario-based Functional Safety Requirements (FSR)

Scenario composition



Scenario-based FSRs

Collision Avoidance Scenarios



GRT AD testing
on lying pedestrian



GRT AD testing — car dummy



GRT AD testing — cyclist

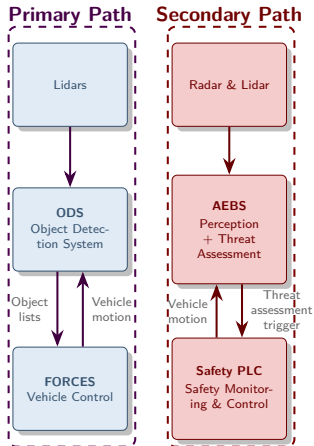


Testing strategy of scenario-based FSRs

- Based on the scenario-based FSRs derived above.
- Each scenario (ego vehicle $\times$ actor type $\times$ actor movement) is validated with a physical test as black-box at vehicle level.
- E.g. put a dummy on the road; if the GRT doesn't hit it, it's compliant.
- Total: **115** scenarios defined (98 single-GRT + 17 GRT-GRT)
- **VRU (Vulnerable Road User) tested:** Infant, Pedestrian, Cyclist, Car, infrastructure etc.
- Effect-chain FSRs are validated via internal log-data.

Effect Chain Safety Requirements

AD architecture overview



Primary effect chains

ODS effect chain

- General object detection functionality
- Gradual response — dynamically calculated braking curve
- Independence from the secondary path

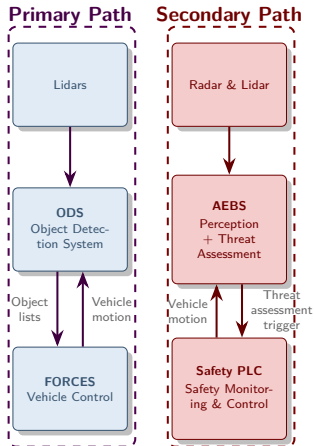
AEBS effect chain

- General collision avoidance functionality
- Irrevocable stop — Time To Collision-based threat assessment
- Independence from the primary path

ISO 26262

Effect Chain Safety Requirements

AD architecture overview



Primary effect chains

ODS effect chain

- General object detection functionality
- Gradual response — dynamically calculated braking curve
- Independence from the secondary path

AEBS effect chain

- General collision avoidance functionality
- Irrevocable stop — Time To Collision-based threat assessment
- Independence from the primary path

ISO 26262

Boundary conditions

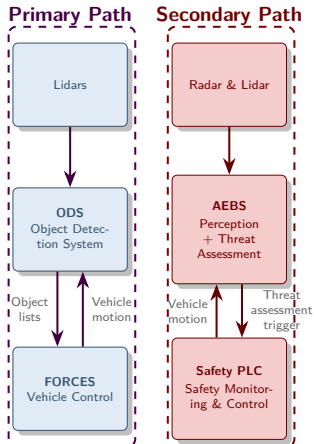
These requirements ensure that the assumptions used in the HARA are satisfied, across six boundary categories:

- Object types
- Lane segregation
- Restricting collision scenarios
- Nominal speed profile
- Occlusion handling

ISO 26262

Effect Chain Safety Requirements

AD architecture overview



Primary effect chains

ODS effect chain

- General object detection functionality
- Gradual response — dynamically calculated braking curve
- Independence from the secondary path

AEBS effect chain

- General collision avoidance functionality
- Irrevocable stop — Time To Collision-based threat assessment
- Independence from the primary path

ISO 26262

Boundary conditions

These requirements ensure that the assumptions used in the HARA are satisfied, across six boundary categories:

- Object types
- Lane segregation
- Restricting collision scenarios
- Nominal speed profile
- Occlusion handling

ISO 26262

Miscellaneous

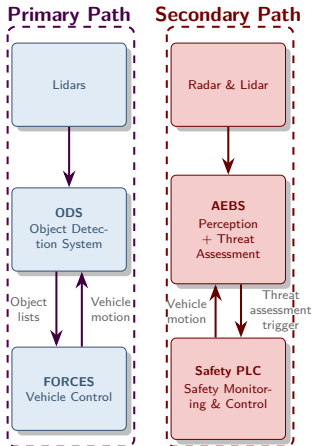
The remaining TSRs — other measures to reduce residual risk.

- Unintended direction.
- Avoid infrastructure triggering.
- Other measures to reduce risks during vehicle passing.

ISO 26262

Effect Chain Safety Requirements

AD architecture overview



Primary effect chains

ODS effect chain

- General object detection functionality
- Gradual response — dynamically calculated braking curve
- Independence from the secondary path

AEBS effect chain

- General collision avoidance functionality
- Irrevocable stop — Time To Collision-based threat assessment
- Independence from the primary path

ISO 26262

Failure detection & handling

- TCA relevant requirements
- ODS and AEBS failure mode detection
- Bypassing feature

SOTIF + ISO 26262

Boundary conditions

These requirements ensure that the assumptions used in the HARA are satisfied, across six boundary categories:

- Object types
- Lane segregation
- Restricting collision scenarios
- Nominal speed profile
- Occlusion handling

ISO 26262

Miscellaneous

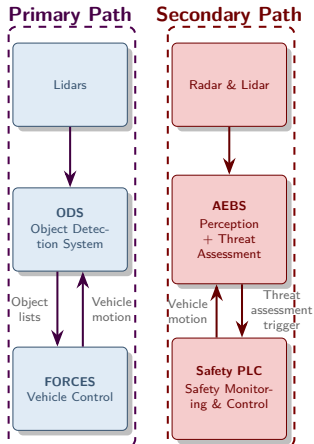
The remaining TSRs — other measures to reduce residual risk.

- Unintended direction.
- Avoid infrastructure triggering.
- Other measures to reduce risks during vehicle passing.

ISO 26262

Effect Chain Safety Requirements

AD architecture overview



Primary effect chains

ODS effect chain

- General object detection functionality
- Gradual response — dynamically calculated braking curve
- Independence from the secondary path

AEBS effect chain

- General collision avoidance functionality
- Irrevocable stop — Time To Collision-based threat assessment
- Independence from the primary path

ISO 26262

Boundary conditions

These requirements ensure that the assumptions used in the HARA are satisfied, across six boundary categories:

- Object types
- Lane segregation
- Restricting collision scenarios
- Nominal speed profile
- Occlusion handling

ISO 26262

Miscellaneous

The remaining TSRs — other measures to reduce residual risk.

- Unintended direction.
- Avoid infrastructure triggering.
- Other measures to reduce risks during vehicle passing.

ISO 26262

Failure detection & handling

- TCA relevant requirements
- ODS and AEBS failure mode detection
- Bypassing feature

SOTIF + ISO 26262

SOTIF-related requirements

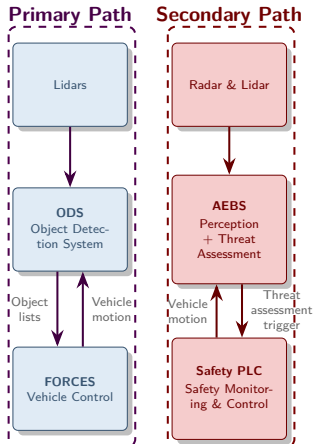
Not required for collision avoidance system, but required for safe operation:

- **Data recording** — vehicle control, external commands from TOMS, ODS, AEBS, CCTV
- **Effects of ODD changes** — preventing ODD changes & start-of-day measures
- **Effects of operating conditions** — environmental/adhesion checks during the operation phase

SOTIF

Effect Chain Safety Requirements

AD architecture overview



Primary effect chains

ODS effect chain

- General object detection functionality
- Gradual response — dynamically calculated braking curve
- Independence from the secondary path

AEBS effect chain

- General collision avoidance functionality
- Irrevocable stop — Time To Collision-based threat assessment
- Independence from the primary path

ISO 26262

Failure detection & handling

- TCA relevant requirements
- ODS and AEBS failure mode detection
- Bypassing feature

SOTIF + ISO 26262

Boundary conditions

These requirements ensure that the assumptions used in the HARA are satisfied, across six boundary categories:

- Object types
- Lane segregation
- Restricting collision scenarios
- Nominal speed profile
- Occlusion handling

ISO 26262

SOTIF-related requirements

Not required for collision avoidance system, but required for safe operation:

- **Data recording** — vehicle control, external commands from TOMS, ODS, AEBS, CCTV
- **Effects of ODD changes** — preventing ODD changes & start-of-day measures
- **Effects of operating conditions** — environmental/adhesion checks during the operation phase

SOTIF

Miscellaneous

The remaining TSRs — other measures to reduce residual risk.

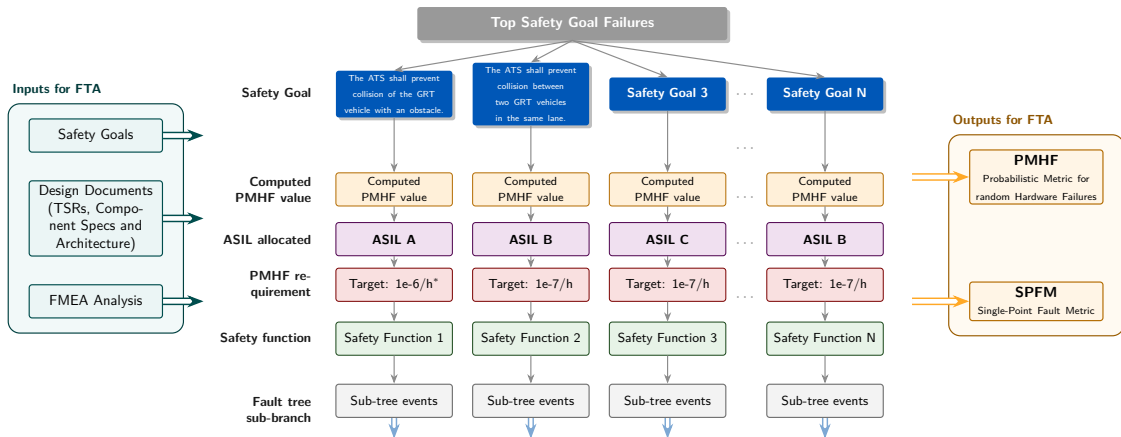
- Unintended direction.
- Avoid infrastructure triggering.
- Other measures to reduce risks during vehicle passing.

ISO 26262

Testing strategy for effect chain

- Effect-chain FSR/TSR are validated via **internal log-data**.
- The input is recorded data.
- The output is comparison of logged parameter values against expected results.
- Applies the Software-level test methods per component.

Quantified Risk Assessment - Fault Tree Analysis



Fault tree structure: from safety goal to basic events

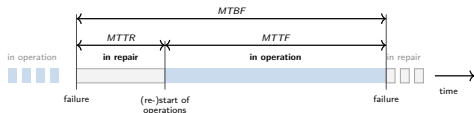
*ISO 26262 does not specify a PMHF target for ASIL A; $1e-6/h$ is a tailored target adopted as common practice within our safety concept.

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

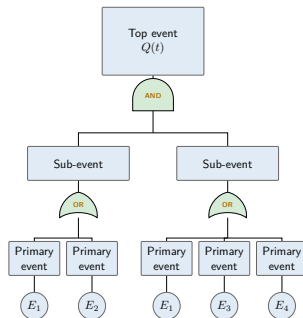
[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262



- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate



Basic-event failure rates λ from FMEA

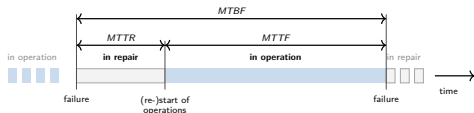
Fault tree construction

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

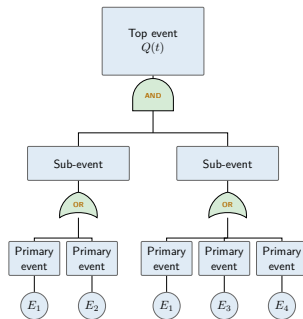
[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262



- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate
- **MTBF** (Mean Time Between Failures):
 $\text{MTBF} = \text{MTTF} + \text{MTTR}$



Basic-event failure rates λ from FMEA

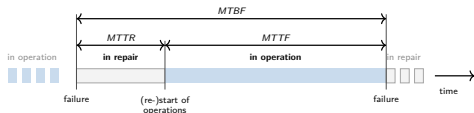
Fault tree construction

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

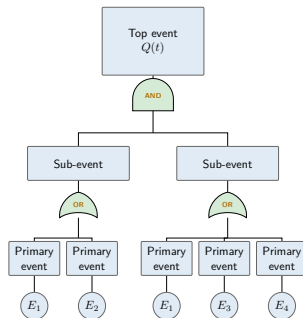
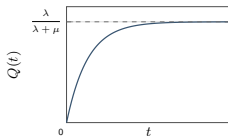
PMHF is not clearly defined in ISO 26262



- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate
- **MTBF** (Mean Time Between Failures):
 $\text{MTBF} = \text{MTTF} + \text{MTTR}$
- **Unavailability**

$$Q(t) := \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

$$\lim_{t \rightarrow \infty} Q(t) = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTR} + \text{MTTF}} = \frac{\text{MTTR}}{\text{MTBF}}$$



Basic-event failure rates λ from FMEA

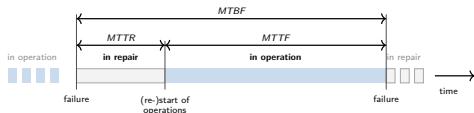
Fault tree construction

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262



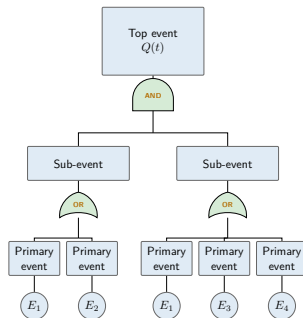
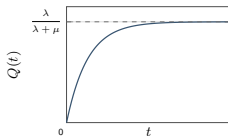
- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate
- **MTBF** (Mean Time Between Failures):

$$\text{MTBF} = \text{MTTF} + \text{MTTR}$$
- **Unavailability**

$$Q(t) := \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

$$\lim_{t \rightarrow \infty} Q(t) = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTR} + \text{MTTF}} = \frac{\text{MTTR}}{\text{MTBF}}$$

- **PMHF** = $\frac{Q(T)}{T}$ — the average failure rate over the operational lifetime T , where $Q(T)$ is the top event's **unavailability** at time T



Basic-event failure rates λ from FMEA

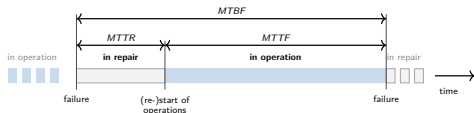
Fault tree construction

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262



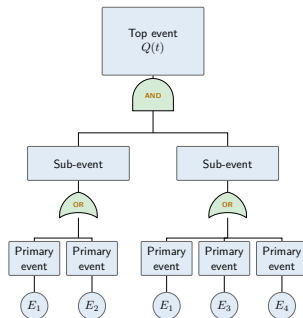
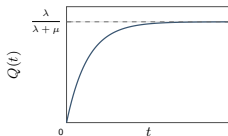
- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate
- **MTBF** (Mean Time Between Failures):

$$\text{MTBF} = \text{MTTF} + \text{MTTR}$$
- **Unavailability**

$$Q(t) := \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

$$\lim_{t \rightarrow \infty} Q(t) = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTR} + \text{MTTF}} = \frac{\text{MTTR}}{\text{MTBF}}$$

- **PMHF** = $\frac{Q(T)}{T}$ — the average failure rate over the operational lifetime T , where $Q(T)$ is the top event's **unavailability** at time T



Basic-event failure rates λ from FMEA

Fault tree construction

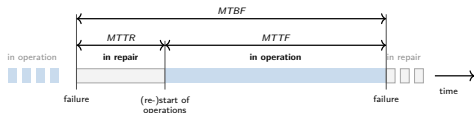
- The complex system is subject to failure as a result of one or more subsystems failing

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262

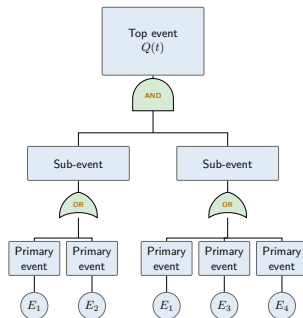
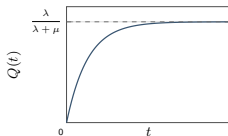


- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate
- **MTBF** (Mean Time Between Failures):
 $\text{MTBF} = \text{MTTF} + \text{MTTR}$
- **Unavailability**

$$Q(t) := \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

$$\lim_{t \rightarrow \infty} Q(t) = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTR} + \text{MTTF}} = \frac{\text{MTTR}}{\text{MTBF}}$$

- **PMHF** = $\frac{Q(T)}{T}$ — the average failure rate over the operational lifetime T , where $Q(T)$ is the top event's **unavailability** at time T



Basic-event failure rates λ from FMEA

Fault tree construction

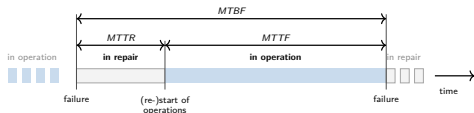
- The complex system is subject to failure as a result of one or more subsystems failing
- Primary event: Component fails

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262



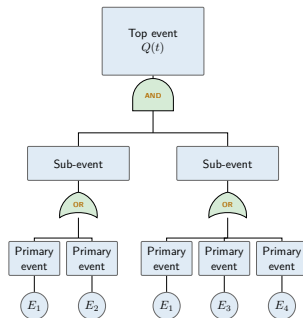
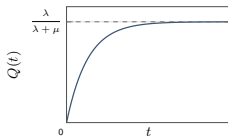
- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the failure rate
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the repair rate
- **MTBF** (Mean Time Between Failures):

$$\text{MTBF} = \text{MTTF} + \text{MTTR}$$
- **Unavailability**

$$Q(t) := \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

$$\lim_{t \rightarrow \infty} Q(t) = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTR} + \text{MTTF}} = \frac{\text{MTTR}}{\text{MTBF}}$$

- **PMHF** = $\frac{Q(T)}{T}$ — the average failure rate over the operational lifetime T , where $Q(T)$ is the top event's unavailability at time T



Basic-event failure rates λ from FMEA

Fault tree construction

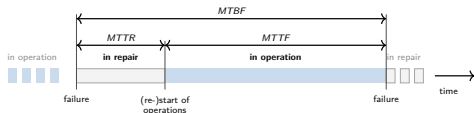
- The complex system is subject to failure as a result of one or more subsystems failing
- Primary event: Component fails
- Sub-event: Subsystem fails

[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Fault Tree Modelling

PMHF is not clearly defined in ISO 26262

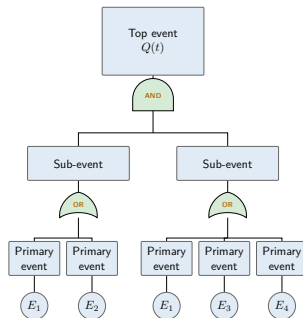
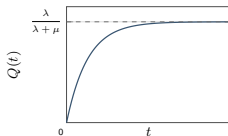


- **MTTF** (Mean Time To Failure) $\lambda := 1/\text{MTTF}$ is the **failure rate**
- **MTTR** (Mean Time To Repair) $\mu := 1/\text{MTTR}$ is the **repair rate**
- **MTBF** (Mean Time Between Failures):
 $\text{MTBF} = \text{MTTF} + \text{MTTR}$
- **Unavailability**

$$Q(t) := \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)t})$$

$$\lim_{t \rightarrow \infty} Q(t) = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTR} + \text{MTTF}} = \frac{\text{MTTR}}{\text{MTBF}}$$

- **PMHF** = $\frac{Q(T)}{T}$ — the average failure rate over the operational lifetime T , where $Q(T)$ is the top event's **unavailability** at time T



Basic-event failure rates λ from FMEA

Fault tree construction

- The complex system is subject to failure as a result of one or more subsystems failing
- Primary event: Component fails
- Sub-event: Subsystem fails
- Top event: Safety goal is violated

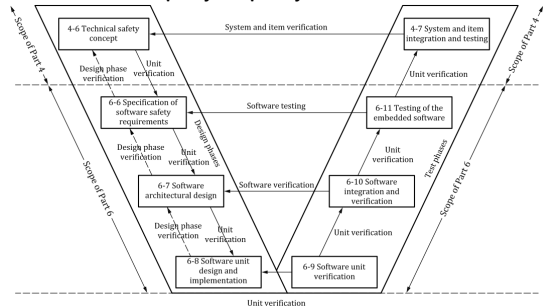
[1] L. Berveling, "Quantified Risk Assessment Report," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1400.001, v3.0, Mar. 17, 2022.

[2] J. Ploeg, "Fault-Tree Analysis: Quantitative Aspects of ISO 26262," 2getthere B.V., internal training presentation, Aug. 6, 2026.

Software Safety

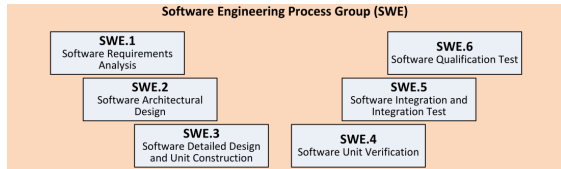
ISO 26262-6 (software product safety) + ASPICE (process capability)

ISO 26262-6 — the quality bar per layer



- Get the inputs from the TSRs, the system architecture and the analysis reports discussed above.

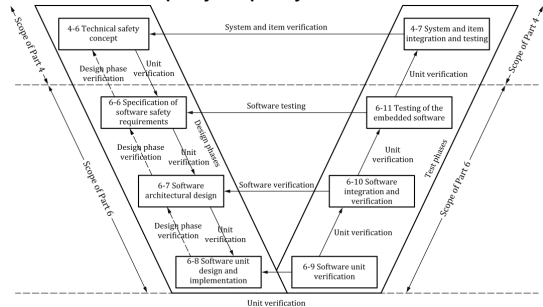
ASPICE — the process skeleton (SWE.1–6)



Software Safety

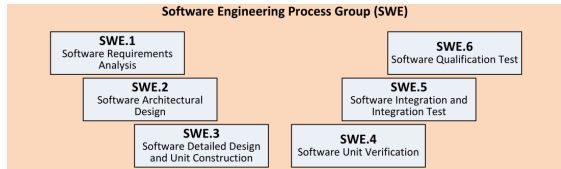
ISO 26262-6 (software product safety) + ASPICE (process capability)

ISO 26262-6 — the quality bar per layer



- Get the inputs from the TSRs, the system architecture and the analysis reports discussed above.
- Specifies the rigour for ASIL A through D at each V-model phase, the **rigour scales, structure does not**.

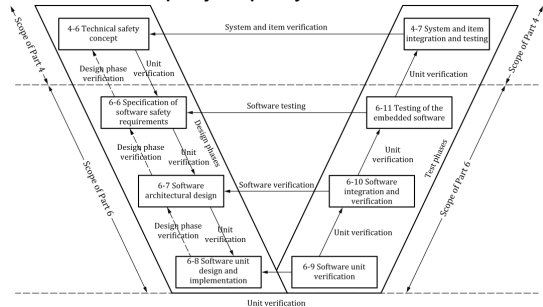
ASPICE — the process skeleton (SWE.1–6)



Software Safety

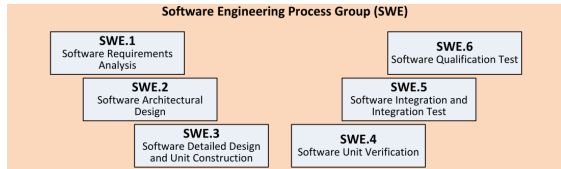
ISO 26262-6 (software product safety) + ASPICE (process capability)

ISO 26262-6 — the quality bar per layer



- Get the inputs from the TSRs, the system architecture and the analysis reports discussed above.
- Specifies the rigour for ASIL A through D at each V-model phase, the **rigour scales, structure does not**.

ASPICE — the process skeleton (SWE.1–6)

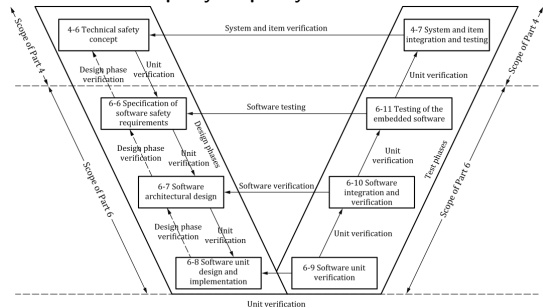


- SWE.1–6: software development phases matching the V-model.

Software Safety

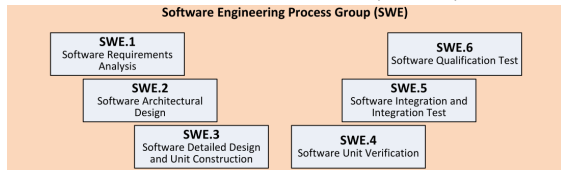
ISO 26262-6 (software product safety) + ASPICE (process capability)

ISO 26262-6 — the quality bar per layer



- Get the inputs from the TSRs, the system architecture and the analysis reports discussed above.
- Specifies the rigour for ASIL A through D at each V-model phase, the **rigour scales, structure does not**.

ASPICE — the process skeleton (SWE.1–6)

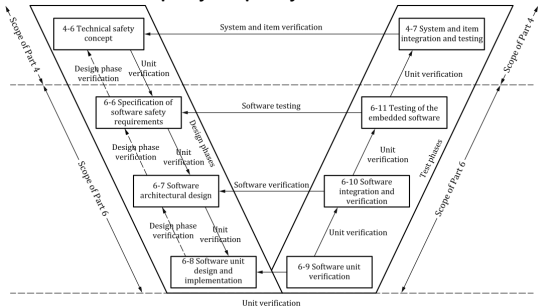


- SWE.1–6: software development phases matching the V-model.
- Defines the lifecycle, makes the work products and the **bidirectional traceability** concrete.

Software Safety

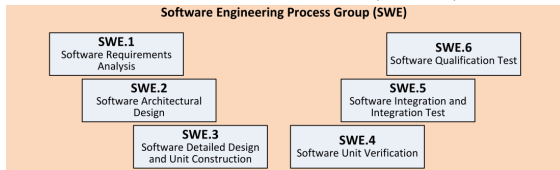
ISO 26262-6 (software product safety) + ASPICE (process capability)

ISO 26262-6 — the quality bar per layer



- Get the inputs from the TSRs, the system architecture and the analysis reports discussed above.
- Specifies the rigour for ASIL A through D at each V-model phase, the **rigour scales, structure does not**.

ASPICE — the process skeleton (SWE.1–6)



- SWE.1–6: software development phases matching the V-model.
- Defines the lifecycle, makes the work products and the **bidirectional traceability** concrete.

Together: efficient ASIL-rated development

ASPICE provides the skeleton; ISO 26262-6 sets the quality bar at each layer. The combination delivers ASIL A/B/C/D work products without re-inventing the lifecycle.

The overlap: ASPICE coverage vs ISO 26262-6 demands

ASPICE = activities + work products + traceability. ISO 26262-6 = technical methods + design principles + ASIL-specific rigour — quantitative where the standard sets one, qualitative otherwise.

Phase	ISO 26262-6 key rigours	ASPICE covers
SW requirements	Must specify: Functional behavior, non-functional constraints, interfaces, configuration parameters. etc.	Software requirement work product, bidirectionally traceable to system-level requirements (TSRs), software architecture and test cases.

The overlap: ASPICE coverage vs ISO 26262-6 demands

ASPICE = activities + work products + traceability. ISO 26262-6 = technical methods + design principles + ASIL-specific rigour — quantitative where the standard sets one, qualitative otherwise.

Phase	ISO 26262-6 key rigours	ASPICE covers
SW requirements	Must specify: Functional behavior, non-functional constraints, interfaces, configuration parameters. etc.	Software requirement work product, bidirectionally traceable to system-level requirements (TSRs), software architecture and test cases.
SW architecture	Control flow and data flow analysis, scheduling analysis, simulation of dynamic behaviour & formal verification. Architecture design principles	Architecture diagram, bidirectionally traceable to SWE.1, SWE.3, SWE.5.

The overlap: ASPICE coverage vs ISO 26262-6 demands

ASPICE = activities + work products + traceability. ISO 26262-6 = technical methods + design principles + ASIL-specific rigour — quantitative where the standard sets one, qualitative otherwise.

Phase	ISO 26262-6 key rigours	ASPICE covers
SW requirements	Must specify: Functional behavior, non-functional constraints, interfaces, configuration parameters. etc.	Software requirement work product, bidirectionally traceable to system-level requirements (TSRs), software architecture and test cases.
SW architecture	Control flow and data flow analysis, scheduling analysis, simulation of dynamic behaviour & formal verification. Architecture design principles	Architecture diagram, bidirectionally traceable to SWE.1, SWE.3, SWE.5.
Detailed design	Design principles. Coding principles mostly covered by MISRA	Detailed design diagram, technical specifications of each unit, bidirectionally traceable to SWE.2 and SWE.4.

The overlap: ASPICE coverage vs ISO 26262-6 demands

ASPICE = activities + work products + traceability. ISO 26262-6 = technical methods + design principles + ASIL-specific rigour — quantitative where the standard sets one, qualitative otherwise.

Phase	ISO 26262-6 key rigours	ASPICE covers
SW requirements	Must specify: Functional behavior, non-functional constraints, interfaces, configuration parameters. etc.	Software requirement work product, bidirectionally traceable to system-level requirements (TSRs), software architecture and test cases.
SW architecture	Control flow and data flow analysis, scheduling analysis, simulation of dynamic behaviour & formal verification. Architecture design principles	Architecture diagram, bidirectionally traceable to SWE.1, SWE.3, SWE.5.
Detailed design	Design principles. Coding principles mostly covered by MISRA	Detailed design diagram, technical specifications of each unit, bidirectionally traceable to SWE.2 and SWE.4.
Unit verification	Static code analysis, coverage scales sharply: statement / branch / MC-DC by ASIL, principles of deriving test cases	Unit test cases, traceable to SWE.3, consistent with SWE.5. Unit test report.

The overlap: ASPICE coverage vs ISO 26262-6 demands

ASPICE = activities + work products + traceability. ISO 26262-6 = technical methods + design principles + ASIL-specific rigour — quantitative where the standard sets one, qualitative otherwise.

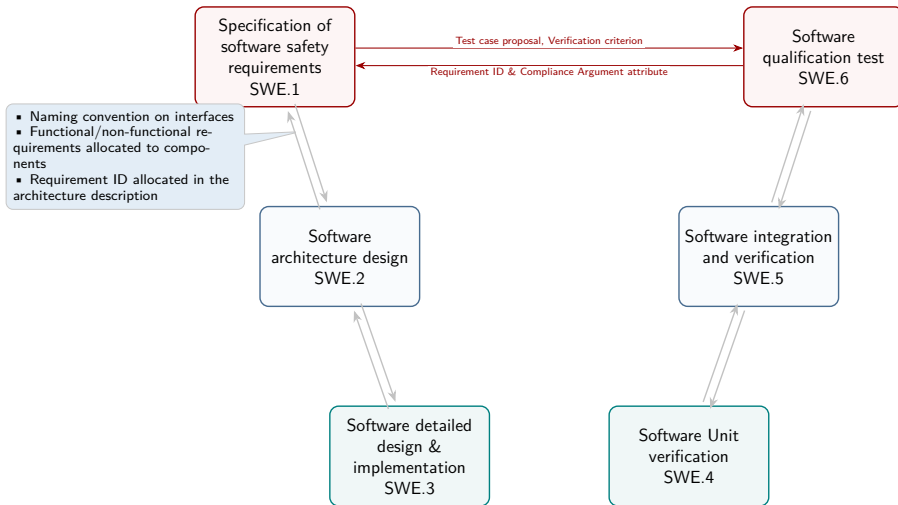
Phase	ISO 26262-6 key rigours	ASPICE covers
SW requirements	Must specify: Functional behavior, non-functional constraints, interfaces, configuration parameters. etc.	Software requirement work product, bidirectionally traceable to system-level requirements (TSRs), software architecture and test cases.
SW architecture	Control flow and data flow analysis, scheduling analysis, simulation of dynamic behaviour & formal verification. Architecture design principles	Architecture diagram, bidirectionally traceable to SWE.1, SWE.3, SWE.5.
Detailed design	Design principles. Coding principles mostly covered by MISRA	Detailed design diagram, technical specifications of each unit, bidirectionally traceable to SWE.2 and SWE.4.
Unit verification	Static code analysis, coverage scales sharply: statement / branch / MC-DC by ASIL, principles of deriving test cases	Unit test cases, traceable to SWE.3, consistent with SWE.5. Unit test report.
Integration test	Function/call coverage scale with ASIL. Error handling, runtime/memory/storage.	Integration-test work product, traceable to SWE.2, consistent with SWE.4

The overlap: ASPICE coverage vs ISO 26262-6 demands

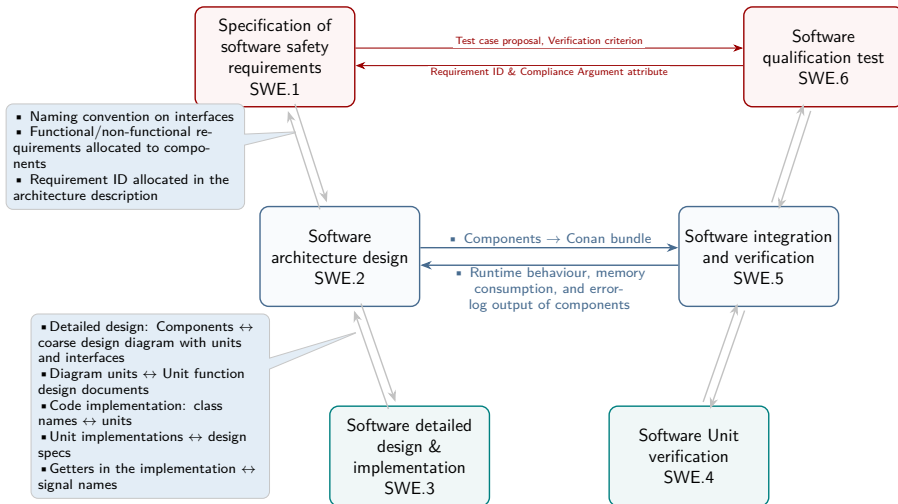
ASPICE = activities + work products + traceability. ISO 26262-6 = technical methods + design principles + ASIL-specific rigour — quantitative where the standard sets one, qualitative otherwise.

Phase	ISO 26262-6 key rigours	ASPICE covers
SW requirements	Must specify: Functional behavior, non-functional constraints, interfaces, configuration parameters. etc.	Software requirement work product, bidirectionally traceable to system-level requirements (TSRs), software architecture and test cases.
SW architecture	Control flow and data flow analysis, scheduling analysis, simulation of dynamic behaviour & formal verification. Architecture design principles	Architecture diagram, bidirectionally traceable to SWE.1, SWE.3, SWE.5.
Detailed design	Design principles. Coding principles mostly covered by MISRA	Detailed design diagram, technical specifications of each unit, bidirectionally traceable to SWE.2 and SWE.4.
Unit verification	Static code analysis, coverage scales sharply: statement / branch / MC-DC by ASIL, principles of deriving test cases	Unit test cases, traceable to SWE.3, consistent with SWE.5. Unit test report.
Integration test	Function/call coverage scale with ASIL. Error handling, runtime/memory/storage.	Integration-test work product, traceable to SWE.2, consistent with SWE.4
Qualification test	HIL testing, vehicle testing, principles of deriving test cases: boundary values/error guessing	Qualification-test work product, traceable to SWE.1

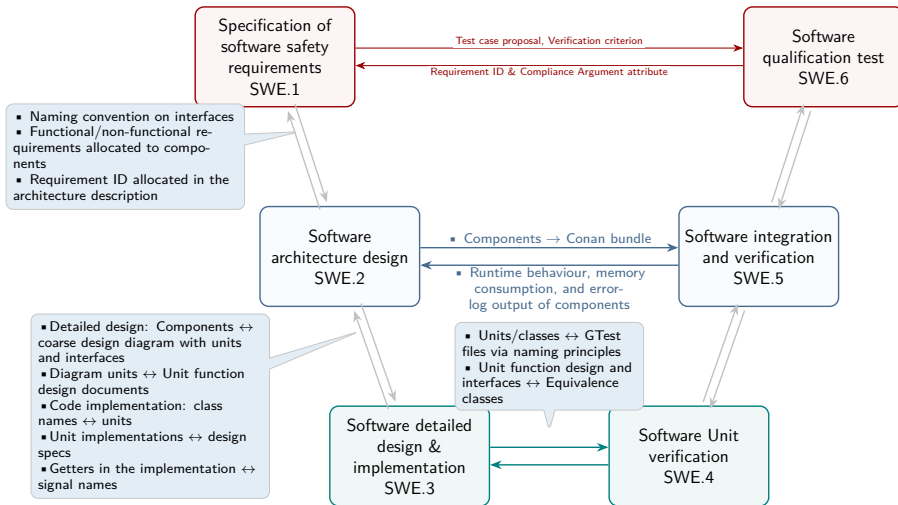
ASPICE — Implementation in the Real World



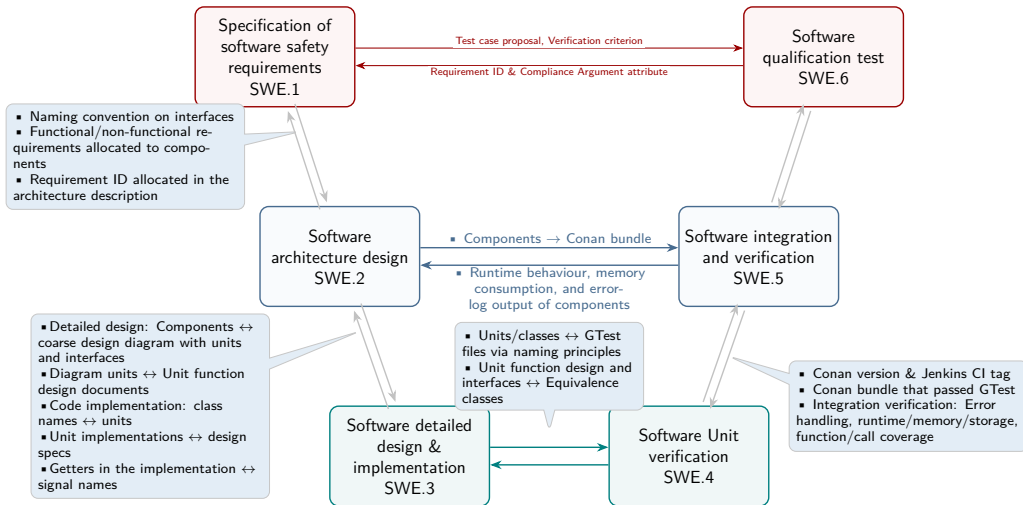
ASPICE — Implementation in the Real World



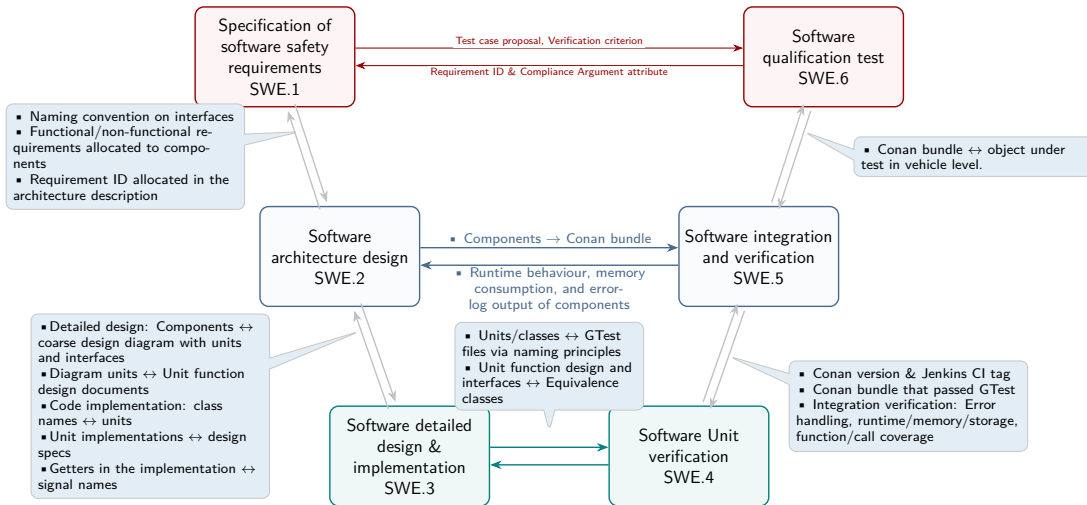
ASPICE — Implementation in the Real World



ASPICE — Implementation in the Real World



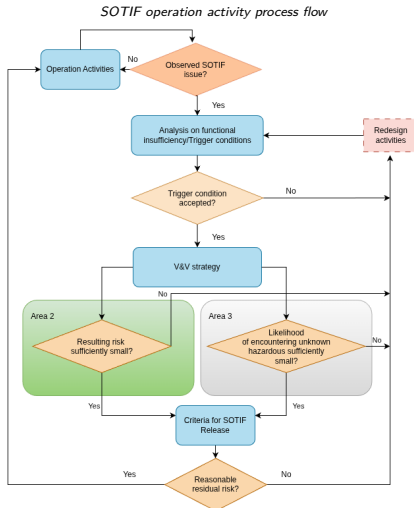
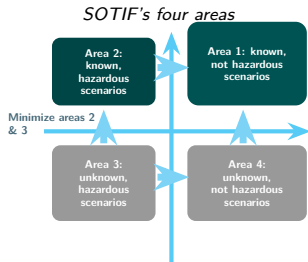
ASPICE — Implementation in the Real World



SOTIF: closing the void where ISO 26262 cannot reach

SOTIF for Collision Avoidance:

Centres on adverse-weather triggering conditions, sensor degradation from the surrounding environment (e.g. sensor interference, retro-reflectors, sensor pollution), and FN/FP performance



Criteria for SOTIF release

- **Risk of FP failure rate acceptable:** FP failure rate within final validation run below x/y km.
- **Risk of FN failure rate acceptable:** Testcases of previous shuttle passed, Testcases from HARA fulfilled.

[1] A. Wohlfarth, "SOTIF Strategy," 2getthere B.V., Utrecht, The Netherlands, Rep. 6011.1100.003, v1.4, Nov. 30, 2022.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.
- **Three-tier safety requirements** with two verification paths: **scenario-based requirements**, verified by scenario/vehicle-level testing, and **effect-chain requirements**, verified by SIL testing with data logging.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.
- **Three-tier safety requirements** with two verification paths: **scenario-based requirements**, verified by scenario/vehicle-level testing, and **effect-chain requirements**, verified by SIL testing with data logging.
- **Hardware failure metrics** required by ISO 26262 are evaluated via **fault tree analysis**.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.
- **Three-tier safety requirements** with two verification paths: **scenario-based requirements**, verified by scenario/vehicle-level testing, and **effect-chain requirements**, verified by SIL testing with data logging.
- **Hardware failure metrics** required by ISO 26262 are evaluated via **fault tree analysis**.
- **Software development:** ASPICE provides the process skeleton; ISO 26262-6 sets the ASIL-specific quality bar at each layer.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.
- **Three-tier safety requirements** with two verification paths: **scenario-based requirements**, verified by scenario/vehicle-level testing, and **effect-chain requirements**, verified by SIL testing with data logging.
- **Hardware failure metrics** required by ISO 26262 are evaluated via **fault tree analysis**.
- **Software development:** ASPICE provides the process skeleton; ISO 26262-6 sets the ASIL-specific quality bar at each layer.
- **SOTIF for collision avoidance** centres on triggering-condition analysis (adverse weather, sensor interference, retro-reflectors, sensor pollution) and FN/FP performance. SOTIF as a whole aims to achieve: unknown-hazardous scenarios become known-hazardous, then are mitigated to known-not-hazardous, by revisiting the design or introducing new measures. This requires operation-phase activities feeding back the operational evidence to drive this loop.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.
- **Three-tier safety requirements** with two verification paths: **scenario-based requirements**, verified by scenario/vehicle-level testing, and **effect-chain requirements**, verified by SIL testing with data logging.
- **Hardware failure metrics** required by ISO 26262 are evaluated via **fault tree analysis**.
- **Software development:** ASPICE provides the process skeleton; ISO 26262-6 sets the ASIL-specific quality bar at each layer.
- **SOTIF for collision avoidance** centres on triggering-condition analysis (adverse weather, sensor interference, retro-reflectors, sensor pollution) and FN/FP performance. SOTIF as a whole aims to achieve: unknown-hazardous scenarios become known-hazardous, then are mitigated to known-not-hazardous, by revisiting the design or introducing new measures. This requires operation-phase activities feeding back the operational evidence to drive this loop.
- **More to come:** our process framework is not limited to the automotive standards track. We are extending coverage to the railway track as well for BKC project in Mumbai, with **EN 50716** bringing SIL-scaled testing rigour, tool qualification, and end-to-end traceability into the same software-development domain.

Conclusions

- **Three-element safety strategy:** our safety strategy for the L4 AD ATS is built on three elements — **ISO 26262** covers E/E functional safety, **SOTIF** covers functional insufficiency, and **ASPICE** covers process assessment.
- **One V-model, three frameworks layered on it:** the whole process follows the ISO 26262 V-model, with SOTIF integrated at the system design and operation phases, and ASPICE supporting the software development phases.
- **Three-tier safety requirements** with two verification paths: **scenario-based requirements**, verified by scenario/vehicle-level testing, and **effect-chain requirements**, verified by SIL testing with data logging.
- **Hardware failure metrics** required by ISO 26262 are evaluated via **fault tree analysis**.
- **Software development:** ASPICE provides the process skeleton; ISO 26262-6 sets the ASIL-specific quality bar at each layer.
- **SOTIF for collision avoidance** centres on triggering-condition analysis (adverse weather, sensor interference, retro-reflectors, sensor pollution) and FN/FP performance. SOTIF as a whole aims to achieve: unknown-hazardous scenarios become known-hazardous, then are mitigated to known-not-hazardous, by revisiting the design or introducing new measures. This requires operation-phase activities feeding back the operational evidence to drive this loop.
- **More to come:** our process framework is not limited to the automotive standards track. We are extending coverage to the railway track as well for BKC project in Mumbai, with **EN 50716** bringing SIL-scaled testing rigour, tool qualification, and end-to-end traceability into the same software-development domain.

Open problems for the community

Statistical assurance for ML perception and continuous certification under retraining remain open beyond ISO 26262 and ISO 21448. We have the operational data; the community has the methods.

Thank you.

Any questions?



Contact

Zonnebaan 1
3542 EA Utrecht
The Netherlands
E-mail: yang@2getthere.eu

